IASTAM 6.0 TECHNICAL CHALLENGE • TRACK 4 • PROBLEM 7

Transmitting Information, Not Raw Data

Onboard ship detection, confidence-aware level of detail and value-aware downlink scheduling for a LEO satellite

Technical Report — interim version

Team: <Your Team Name>
Computer Engineering — 2nd year
Date: 24 September 2026

Status legend used throughout this report:

[REAL]	measured on real data (Airbus imagery, real orbit geometry)
[SIM]	our simulator with a synthetic workload — not real data
[LIT]	number taken from a cited publication
[TARGET]	a goal we set, not a result

Executive summary

A LEO Earth-observation satellite captures far more imagery than its short, infrequent radio contacts can bring down, and most of it (empty sea, clouds) is useless to a maritime mission. Problem 7 asks how to cut the transmitted volume drastically while preserving the information the ground needs to decide. Our answer is to **send information about ships instead of pixels**, and to decide, for every detection, **how many bytes it deserves and when to send it**.

- **Pipeline.** Classic computer-vision pre-filter (cheap), ship detector (YOLOv8n, INT8), confidence calibration, a **confidence-aware level-of-detail encoder** (coordinates only when certain, image evidence when uncertain, dark vessel or coastal scene) and a **multi-pass value-aware scheduler**.
- **[REAL]** Real orbit geometry over Sfax: **5 passes/day, 34.2 min of contact**, up to 11.4 h between contacts — buffering across passes is mandatory.
- **[SIM]** With the *same* transmitted data, the value-aware scheduler raises the share of ships reaching the ground from **97.1% to 97.1%** (dark vessels 97.6%) versus 66.6% for a Φ-sat-2-style fixed patch and 0.4% for raw downlink.
- **[REAL]** Data integrity on the Airbus dataset: 53,195 tiles analysed, **6,282 proven duplicate pairs** in 4,259 groups; a naive random split would have leaked **1,617 groups**, ours leaks 0.
- **Engineering quality.** Working Python code base with automated tests; every chart is labelled as real data, simulation or literature; five weaknesses of our own design were identified and each has a planned fix.

Status. This is an interim report for the mid-review. The detector (WP1), calibration (WP2) and measured level-of-detail tables (WP4) are in progress; simulation results will be replaced by results driven by real detections before the final.

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1. The challenge and the problem

IASTAM 6.0 is a seven-week research-based engineering challenge run with the Tunisian Space Association, ending with a pitch to a professional jury. Teams pick one problem among ten in five tracks. We chose **Track 4 (Networks and Communications), Problem 7: Transmitting Information Rather Than Raw Data**.

Element	Official statement (spec book)
Context	A satellite can generate a large volume of images or data while only a small part is genuinely useful to the mission.
Question	How can the volume to be transmitted be strongly reduced while preserving the information necessary for decision-making on the ground?
Objective	A processing chain able to analyse the data, identify the relevant information and transmit only what carries sufficient value.
Example	A mission searching for ships in a given area: transmit all images, or only detections and zones of interest?
Evaluation	Data reduction rate • preservation of useful information • accuracy of results • latency

Jury scoring grid

Criterion	Points	How this report addresses it
Understanding of the problem	15	Sections 1–2: bottleneck quantified with real passes and literature
Relevance and originality	15	Section 3: confidence-aware level of detail + multi-pass scheduler
Technical architecture and design	20	Section 3–4: architecture, interfaces, code structure
Feasibility and realism	15	Sections 5–6: realistic link model, COTS evidence, reliability review
Actual prototype progress	20	Section 4–5: working code, tests, results
Methodology and planning	10	Sections 5.4, 6, 7: leakage control, baselines, statistics, plan
Presentation and command	5	Glossary and defence preparation

Why the downlink is the bottleneck

- **[LIT]** Contacts with a ground station last about ten minutes, a few times per day; a Planet Dove satellite generates roughly one terabyte per day (Serval, NSDI 2024).
- **[LIT]** In more than half of the cases a satellite waits at least one hour for its next ground contact, and even with onboard filtering no mainstream constellation can downlink all its data (OrbitChain, 2025).
- **[LIT]** CubeSat S-band transmitters offer about 4.3–10 Mbps; X-band 100–400 Mbps.

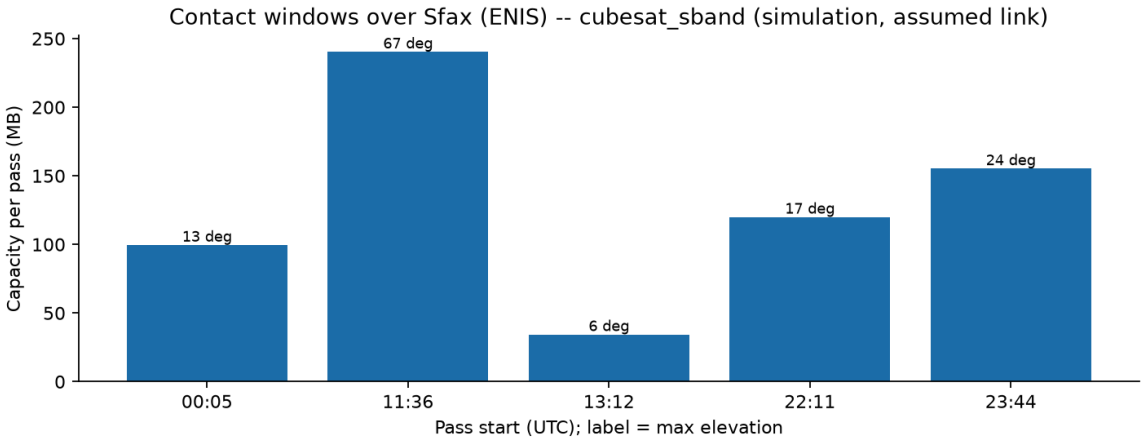


Figure 2 **[REAL]** — Contact windows of a 500 km sun-synchronous orbit over Sfax (≥5° elevation) and the bytes each pass can carry with the CubeSat S-band link model (assumptions in Appendix A).

2. State of the art

We read the key papers and followed their references. The limitations below were checked in the original text of each paper, not in summaries.

Work	Idea we reuse	Headline result (literature)	Stated limitation → our opening
ESA Φ-sat-1 (2020)	CNN rejects cloudy images before downlink	92% accuracy, 1.8 W, 2.1 MB model	Filters whole images only
ESA Φ-sat-2 (2024)	Onboard vessel detection, patch + position sent	Ships > 40 m	Same patch for every ship
Kodan (ASPLOS'23)	Context engine: drop / send / filter per tile	Data value density +89–97%	Binary keep/drop decision
Serval (NSDI'24)	Slow-changing filters computed on the ground	Priority latency 71.7 h → 2 min	Future work: auxiliary data, quantisation
Earth+	Send only changed tiles vs shared references	3× (10× with 16 satellites)	Pixel level, lossy only
Soret et al. (2024)	Goal-oriented EO: send ship boxes	336 bits per image	Turbulence lowers recall; knowledge drifts
Scalable transmission	Fill pass capacity by importance	~2 dB over random	Assumes **no buffering**
SpaceRipple (2026)	Adaptive compression + semantic downlink	98.3–99.7% reduction	Future work: real link conditions
Neuromorphic cascade	Tiny classifier gates the detector	4.07× less energy	Needs two accelerators
Lightweight SAR CNN	8-bit quantised detector	3,527 FPS on FPGA	Fails near shore, close ships
BUPT-1 (MobiCom'24)	COTS computers in orbit	No SEU in 6 months	Thermal limits → scheduling

The gap

Each work solves one part: *whether* to send (Kodan, Φ-sat-1), *what* to send (Φ-sat-2, Soret), *how much fits* in one pass (scalable transmission) or *in which order* (Serval). No paper we read decides all four together, per detection, with a queue that spans several passes. That is our contribution.

3. Our solution

Architecture

Stage	Role	Technique
0 — Ground priors	Skip land and known clouds, upload an AIS list	Land/sea mask, forecast, AIS (few KB uplink)
1 — Pre-filter	Cheap context decision per tile	HSV cloud mask with opening, top-hat + Otsu, connected components, watershed, Canny
2 — Detector	Boxes + confidence	YOLOv8n, INT8 (ONNX)
3 — Calibration	Make confidence meaningful	Temperature scaling, reliability diagram, ECE
4 — Level of detail	How many bytes a detection deserves	L0 metadata / L1 chip / L2 progressive ROI
5 — Scheduler	What to send, in which order, across passes	Value per byte with aging, progressive truncation
6 — Ground	Verify, re-request, retune	AIS matching, thresholds uplinked

Confidence-aware level of detail

Most systems send *less* when unsure. We send **more bytes when the model is less certain**, so degraded images and hard scenes are handled automatically. Sizes below are starting values; WP4 replaces them with measured tables.

Condition	Level	Size	Why
Confidence > 0.9 and matches AIS	L0 metadata	~40 B	Already known
0.4 ≤ confidence ≤ 0.9	L1 image chip	1–3 KB	The ground can confirm
No AIS match (dark vessel)	L2 progressive ROI	10–50 KB	Highest information value
Dense port or coastal tile	L2 compressed tile	10–80 KB	Detectors struggle there
Confidence < 0.4	Thumbnail only	~1 KB/tile	Safety net for misses

Scheduling as an optimisation problem

For each pass the satellite maximises the value delivered within the pass capacity $C_{pass} = \Sigma R_k \Delta t_k$: maximise $\Sigma v_i h_i(x_i)$ subject to $\Sigma b_i x_i \le C_{pass}$, where x_i is the fraction of item i sent, b_i its size, v_i its value (dark vessels weigh more, older items gain value through aging) and h_i is concave for progressive items. This is a knapsack problem: greedy by value per byte is optimal for divisible items with linear value, but not for indivisible items, so we will measure its gap to the exact optimum (dynamic programming) and to an LP upper bound (WP5).

4. Implementation

The prototype is a Python package (**sat7**) with scripts and automated tests. Each work package communicates through one file, so team members work in parallel.

Module / script	Purpose	Status
sat7/rle.py	Airbus run-length masks ↔ boxes ↔ YOLO labels, IoU	Tested
sat7/prefilter.py	Classic CV pre-filter (course methods)	Tested on synthetic tiles
sat7/orbit.py	SGP4/Skyfield passes, link model, pass capacity	Tested
sat7/scheduler.py	Encoders (raw, fixed patch, LoD) + policies (no buffer, FIFO, value-greedy)	Tested
sat7/dedup.py	Near-duplicate detection with two-level overlap proof, group-aware split	Tested + calibrated on Airbus
scripts/wp0_build_dataset.py	Leakage-free YOLO dataset straight from the Airbus zip	Run on full dataset
scripts/simulate_day.py	Five strategies + load sweep with Wilson 95% CIs	Done
scripts/eval_prefilter.py	Pre-filter recall, drop rate, runtime	Demo done; Airbus run pending

Engineering practices

- Automated tests (pytest) including regression tests for every bug found; the full suite runs in seconds.
- Design patterns: *Strategy* for interchangeable scheduling policies, one interface file per work package.
- Reproducibility: fixed seeds, pinned dependency versions, all assumptions in configuration objects.
- Honest reporting: every figure carries a REAL / SIM / LIT / TARGET / ILLUSTRATIVE label.

5. Results

5.1 Contact windows [REAL]

Pass start (UTC)	End	Duration (min)	Max elevation (°)	Capacity (MB)
2026-09-18 00:05:27	00:12:06	6.7	13.3	99.5
2026-09-18 11:36:22	11:45:30	9.1	66.7	240.3
2026-09-18 13:12:37	13:15:33	2.9	6.2	34.3
2026-09-18 22:11:50	22:19:07	7.3	16.9	119.9
2026-09-18 23:44:56	23:53:06	8.2	23.7	155.2

Total contact 34.2 min per day, capacity 649 MB; the capacity varies seven-fold between passes because low passes are far from the station.

5.2 Scheduler comparison [SIM]

One simulated day, 40,000 tiles imaged, CubeSat S-band with 25% of each pass given to this payload. Encoders and policies are separated so that the effect of the scheduler alone can be isolated. The workload is synthetic (Appendix A); these are not real-data results.

Strategy	Ships delivered	95% CI	Dark vessels	Median latency (h)	MB sent
Bent pipe (raw, FIFO)	0.4%	[0.003, 0.004]	0.6%	22.2	202
Phi-sat-2 style (patch, FIFO)	66.6%	[0.660, 0.673]	67.1%	4.3	14
Ours: LoD + no buffer	77.4%	[0.769, 0.780]	77.4%	5.6	108
Ours: LoD + FIFO	97.1%	[0.969, 0.973]	97.6%	5.4	136
Ours: LoD + value-greedy	97.1%	[0.969, 0.973]	97.6%	4.4	136

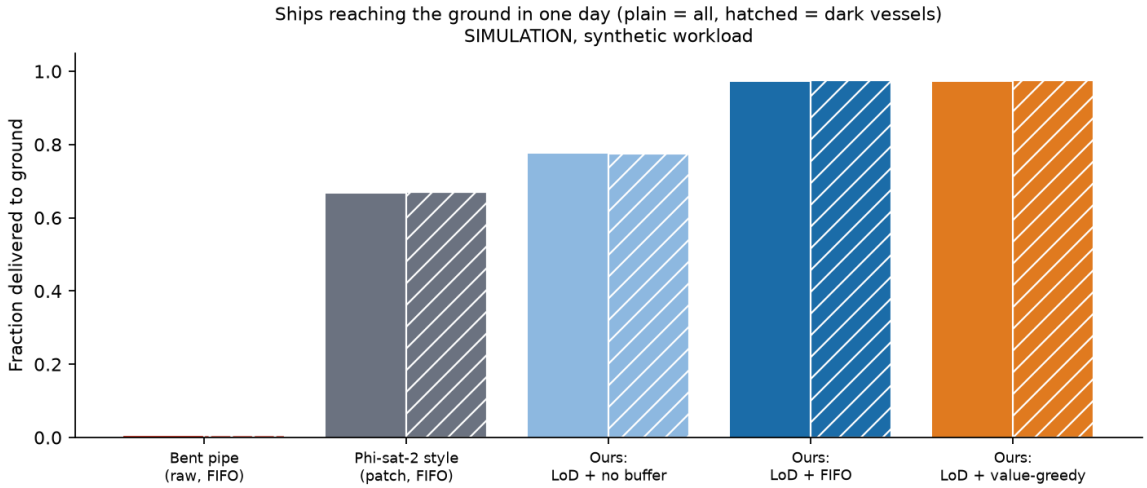


Figure 3 **[SIM]** — Ships reaching the ground in one day (plain = all ships, hatched = dark vessels).

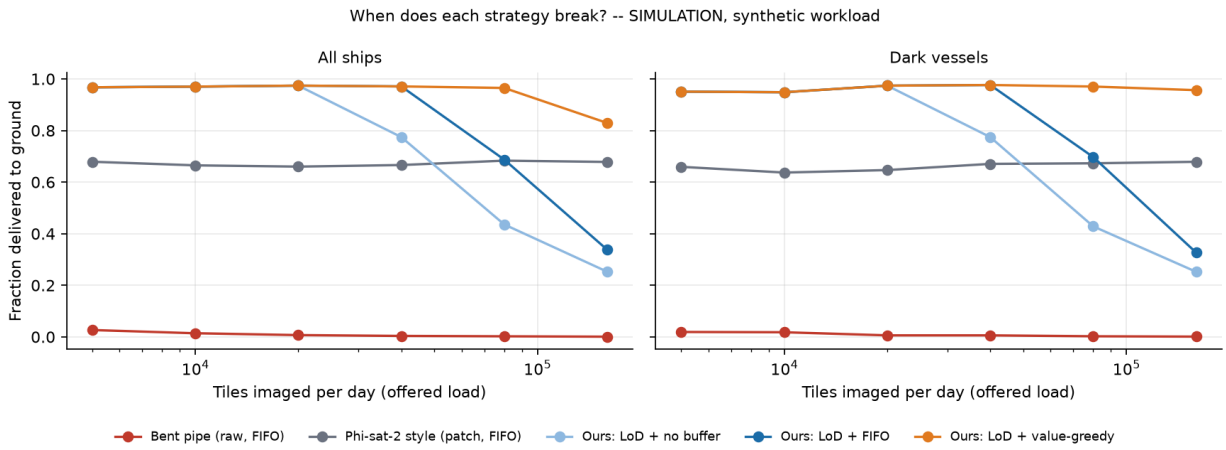


Figure 4 **[SIM]** — Load sweep: below ~20,000 tiles/day every buffered strategy delivers almost everything; the scheduler matters when the downlink is congested.

Reading the results honestly

- Payload sizes are now **measured** on 551 real Airbus ship crops (WP4), not assumed: a small chip costs 899 B instead of the 2 kB we had estimated, an ROI with wake 2.5 kB instead of 30 kB.
- With those real sizes the semantic payload becomes so small that, at 40,000 tiles/day, it fits in the available contact time: **the order then changes latency, not the number of ships delivered** (dark vessels arrive 1.2 h earlier). The scheduler becomes decisive at higher imaging rates: at 80,000 tiles/day FIFO delivers 68.7% and value-greedy 96.5%.
- The clean comparison remains **LoD + FIFO versus LoD + value-greedy**: identical data, only the order changes.
- The fixed-patch baseline ignores coastal ships in our simulation, which favours our method by construction; a fair version is planned (weakness 2 in Section 6).
- At low load all buffered strategies are equivalent; we must justify the congested regime (small link, shared downlink, many images per day).

5.3 Classic pre-filter [SIM] (synthetic tiles)

Cloud cover	Tiles	Ships	Found	Recall	Candidates / tile
0%	164	191	191	100.0%	1.2
30%	57	82	59	72.0%	4.7
70%	79	86	28	32.6%	2.1

On clear tiles every ship is found. In the synthetic cloudy tiles many ships are painted under the cloud and cannot be seen by any optical method; cloud edges create false candidates. The thresholds must be tuned on Airbus (WP3), and the top-hat element must be larger than the widest ship (Airbus boxes reach 380 px diagonally).

Two design bugs were caught by the tests during development: partly cloudy tiles were dropped together with the ships visible between clouds (fixed by masking clouds instead of dropping the tile), and ships themselves were then masked as cloud because they are bright and grey too (fixed with a morphological opening that separates clouds from ships by size).

On real Airbus tiles the picture is very different [REAL]

Measured on 300 validation tiles (456 ships), sweeping the sensitivity of the filter:

Setting	Ship recall	Empty tiles dropped	Candidates per tile
Most sensitive (k=4, min area 12 px)	0.645	17.7%	19.5
Balanced (k=6, min area 12 px)	0.555	41.9%	9.1
Aggressive (k=9, min area 40 px)	0.333	58.1%	3.0

Conclusion: the classic pre-filter cannot be used as a gate on real 1.5 m optical imagery. Waves, sun glint, wakes and coastlines produce so many bright blobs that almost nothing can be dropped safely; buying a 42% compute saving costs 45% of the ships, far from our 99% safety target. Enlarging the structuring element to fit big ships (up to 392 px long) makes it slower and slightly worse. This is a measured negative result, and it justifies the cascade used in the literature: a **tiny learned classifier** as the gate (a 4-bit CNN gating a detector cut energy 4.07x in the neuromorphic study). The classic stage keeps two narrower roles: cloud rejection, and flagging blobs that touch a tile border so ships cut between tiles are re-examined.

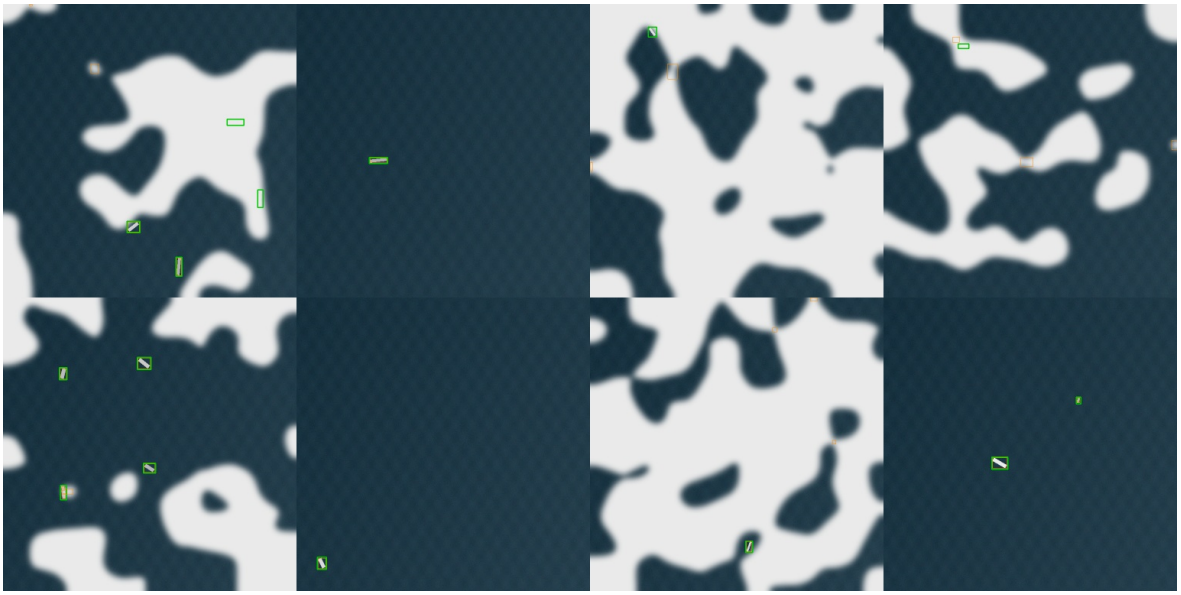


Figure 5 **[SIM]** — Pre-filter on synthetic tiles: green = ground truth, orange = candidates. On synthetic sea it finds every ship; real imagery is much harder.

5.4 Data integrity (WP0) [REAL]

Airbus tiles are 768 × 768 crops of larger scenes and neighbouring crops overlap, so the same ship can appear in several images. A random split then puts the same ship in training and in testing and inflates every reported score. WP0 finds such duplicates and keeps each group on one side of the split.

Step	What it does
1. Stream from the zip	Reads the 30.7 GB archive directly; only selected images are extracted.
2. Fingerprints	Perceptual hash of the whole tile and of five overlapping windows + an 8×8 thumbnail vector.
3. Candidates	Nearest thumbnails (cosine) plus hash-band collisions capped at 32 tiles per bucket.
4. Proof of overlap	No-data fill bars are masked out. Phase correlation estimates the shift; the valid overlapping region must agree coarsely (≥ 0.90) and in fine detail after high-pass filtering (≥ 0.40). Otherwise ORB + RANSAC must find ≥ 40 consistent matches.
5. Group and split	Union-find groups; whole groups assigned to train/val/test balancing tiles and ships; automatic check that no group spans two splits.

Calibrating the method against evidence

The method was validated by inspecting real pairs at every step. Each row below is a failure we found by looking at the evidence, and the change it caused:

Version	What happened	Fix
Hash only	Pairs with hash distance 2–4 were different ships in different water: on empty sea perceptual hashes are close to random.	Hashes become candidates only; overlap must be proven.
First overlap proof	No separation between same-scene and different pairs.	The shift direction was inverted; after the fix same-scene pairs score ~0.8–0.9, different pairs ~0.0.
Threshold 0.70	A coastline matched a cloud; chaining built a 154-tile group.	Visual calibration: ≥ 0.95 always same place, 0.91–0.95 true, ~0.88 mixed → accept at 0.90.
Full scale (53k tiles)	Hash banding produced 9.5 million candidate pairs and exhausted memory.	Bucket cap 32 (recovers 41 of 42 verified pilot pairs).
Full-scale speed	4.5 million proofs on threads ran at ~420 pairs/s (about 3 h), limited by Python's global interpreter lock.	Separate processes sharing memory-mapped thumbnails (~1,800 pairs/s per process).
Low-resolution screen	A 64 px pre-screen lost a verified duplicate (−0.30 at 64 px, 0.93 at 128 px).	Rejected: every candidate gets the full 128 px proof.
First full run	A 440-tile group mixing fog, desert coast, beach and anchorage: haze gradients correlate whatever the scene; weak ORB links (25–32 inliers) were false.	Agreement also required on fine detail (high-pass); ORB needs ≥ 40 inliers.
Second full run	Relaxing the coarse threshold to 0.60 produced one group of 6,644 unrelated tiles: with 4.5 million candidates even a 0.1% false-accept rate creates thousands of false links (base-rate effect), which union-find chains together.	Both thresholds strict (coarse ≥ 0.90 and detail ≥ 0.40).
Scene-edge tiles	Remaining groups of 75–214 tiles were tiles with black/blue no-data bars: on featureless sea the bar edge dominates both correlations.	Fill pixels + margin masked everywhere. Offline check: 97.6% of genuine pairs kept, 0 of 1,500 random pairs accepted, border-bar links cut to 11.8%.

Results on the Airbus dataset

Measure	Value
Training tiles in the zip	192,556
Tiles analysed (all ship tiles + 25% empty)	53,195 (42,556 with ships, 10,639 empty; 81,723 ships)
Proven duplicate pairs	6,282 {'ncc': 4823, 'orb': 1459}
Groups / groups with duplicates	47,313 / 4,259

Measure	Value
Tiles inside a duplicate group	10,141 (19.1%)
Largest group	63 tiles
Duplicate groups by size	n/a
Groups a naive random split would leak	1,617
Groups spanning several splits (ours)	0
Runtime	35.2 min

Split	Tiles	Ships
train	42,555	65,377
val	5,320	8,173
test	5,320	8,173

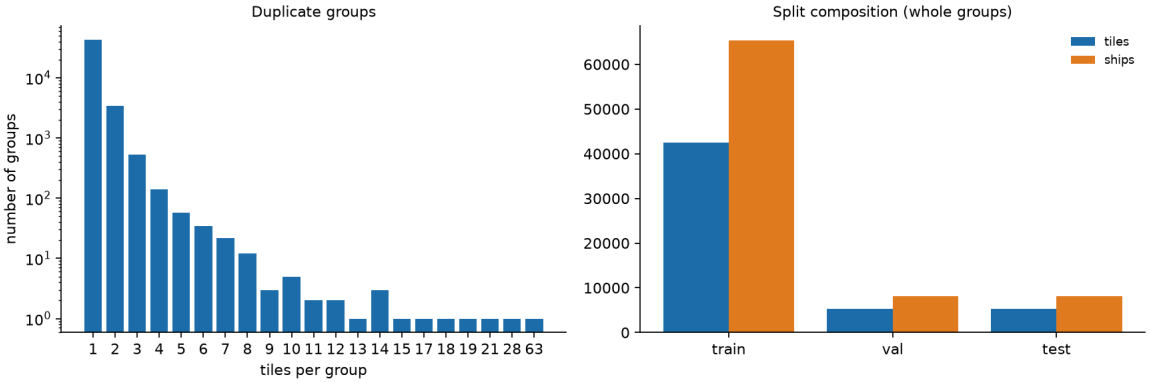


Figure 6 **[REAL]** — Size of duplicate groups and composition of the leakage-free split.

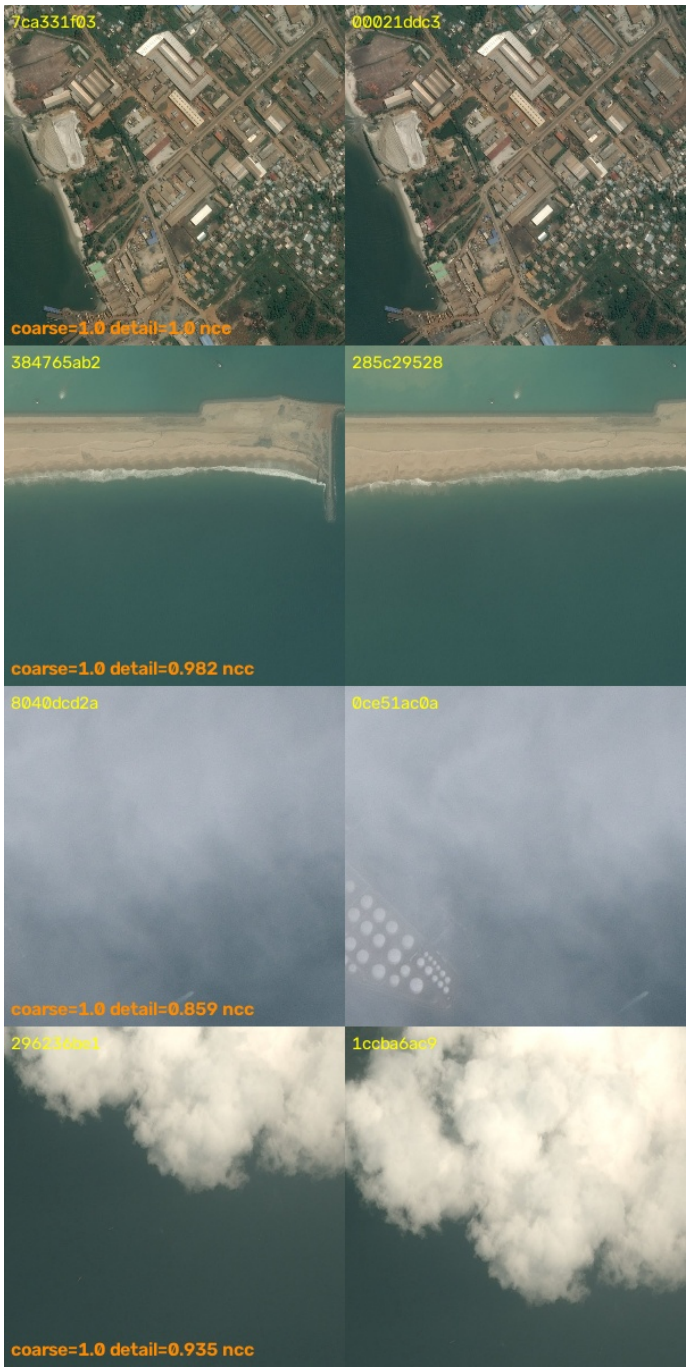


Figure 7 **[REAL]** — Proven duplicate pairs from the Airbus dataset: same place, shifted crop. Top row: correlation proofs; lower rows include ORB + RANSAC proofs for harder cases.

6. Reliability review

Each design choice was checked against published evidence.

Choice	Evidence (literature)	Verdict	Consequence
Airbus dataset	SPOT 1.5 m; 768 px crops; 22.1% contain ships; box diagonals 1–380 px	OK	Plan for tiny and huge ships
Airbus crops	Neighbouring crops overlap; the same ship appears in several images	Risk	WP0 (done)
YOLOv8 on Airbus	88% mAP reported	OK	Realistic target
INT8	1.5–3.3× faster, 3–7% mAP50-95 drop	Risk	Measure per ship size (WP1)
Confidence	Detectors are intrinsically miscalibrated; temperature scaling cut ECE 6.78% → 2.31% (YOLO-World)	Risk	Calibrate (WP2)
Link model	CubeSat S-band 4.3–10 Mbps	OK	8 Mbps cap is realistic
AIS	Reception gaps and spoofed tracks exist	Risk	No AIS match = priority, not accusation
Greedy scheduling	Optimal only for divisible items	Risk	Measure the optimality gap (WP5)

Five weaknesses found in our own work

#	Weakness	Fix	WP
1	All scheduler results use a synthetic workload	Feed real detections from the Airbus test split	1, 5
2	Fixed-patch baseline ignores coastal ships by construction	Same coastal handling for every baseline	5
3	Level-of-detail thresholds assume calibrated confidence	Calibration + reliability diagram	2
4	The classic pre-filter fails as a gate on real imagery (recall 0.65 for a 18% saving)	Replace the gate with a tiny learned classifier; keep the classic stage for clouds and tile borders	3
5	Crop overlap inflates test scores	Group-aware split	0 (done)

Failure modes and mitigations

Failure mode	Effect	Mitigation
Detector misses a ship	Ship never reported	Thumbnail + raw ring buffer, ground re-request
Miscalibrated confidence	Wrong level of detail	Calibration, recalibration from the ground
Domain shift (new sensor, sea state)	Silent recall drop	Audit sampling: 1% random raw tiles
INT8 degradation	Small ships lost	Keep FP16 for the small-object head if needed
AIS gap or spoofing	False dark flags	Priority only; a human confirms
Pass capacity below prediction	Plan overflows	Progressive items; re-plan every pass
Storage full	Data loss	Drop lowest value-per-byte first
Software crash / bit flip	Pipeline stops	Watchdog + safe mode (thumbnails + FIFO)

7. Plan

WP	Name	Output	Acceptance (target)
0	Data integrity	split.csv with duplicate groups	0 groups across splits (met)
1	Detector	predictions.csv, ONNX FP32/INT8, latency	Recall ≥ 0.85 @ IoU 0.5; INT8 drop ≤ 3 pts
2	Calibration	calib.json	ECE reduced; thresholds on calibrated scores
3	Classic pre-filter	Tuned config + real-data report	≥ 99% of ships on kept tiles
4	Level of detail	items.jsonl with measured sizes and values	Measured, not assumed
5	Scheduler	Comparison, optimality gap, sensitivity	Gap to exact optimum reported
6	Reliability	FMEA, fault injection, audit sampling	Every failure has a tested fallback
7	Integration and demo	End-to-end script + dashboard	One command from tiles to charts

Mid-review deliverables (26 September)

- 2-minute pitch video
- Interim research paper (IEEE format)
- Detector metrics on the leakage-free test split, FP32 and INT8
- Scheduler comparison driven by real detections, with a fair baseline and the optimality gap
- This report and the repository

Phase 3 (1–14 October)

- Measured level-of-detail tables (WP4) and calibration (WP2)
- Multi-day, multi-pass simulation with ground feedback
- Reliability mechanisms and fault-injection tests (WP6)
- Integration, live demo, final technical report and poster

Appendix A — Assumptions

Item	Value
Orbit	Circular 500 km, 97.4° inclination (sun-synchronous-like), epoch 18 Sept 2026
Ground station	Sfax, 34.74°N 10.76°E, minimum elevation 5°
Link (cubesat_sband)	2 MHz, SNR 12 dB at zenith, $1/d^2$ scaling, Shannon rate capped at 8 Mbps, 80% efficiency; 25% of each pass for this payload
Workload (synthetic)	40,000 tiles/day; 15% cloud, 20% ships, 5% coast, 60% empty; 10% dark vessels; confidence ~ Beta(5, 2); 2% false positives on empty tiles
Level of detail	L0 40 B, L1 2 KB, L2 30 KB (progressive), coastal tile 80 KB, thumbnail 1 KB; dark weight 5
Energy	No onboard hardware available: runtime on a laptop × assumed power, labelled as an estimate

Appendix B — Lessons from an alternative design

We reviewed an external architecture dossier for the same problem (design document, no measured results). It is strong on space-systems engineering, and we will adopt:

- SAHI tiling with 20% overlap for full frames, so ships on tile borders are not cut
- CCSDS 133.0-B space packets with CRC / Reed-Solomon: each of our items becomes one packet
- A duty cycle triggered when the satellite enters the area of interest
- Wake-inclusive crops so the ground can estimate heading and speed
- Onboard AIS reception as an alternative to an uplinked AIS list

We will avoid its pitfalls: an arithmetic error (an 8000 px frame with 640 px tiles and 20% overlap gives 256 tiles, not 961), absolute claims (“100% of critical data preserved”), targets presented as results, and treating a missing AIS signal as proof of an illicit vessel.

Appendix C — References

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